They Like to Move It, Move It — Determining the Viscosity of Colloidal Suspensions of Monodisperse Polystyrene Microspheres in Water Using Brownian Motion

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Abstract

I. INTRODUCTION

Brownian motion describes the continuous, random movement of microscopic particles suspended in a fluid. The phenomenon was first observed by Robert Brown in the nineteenth century, who noticed the irregular movement of pollen grains suspended in water. Its theoretical foundations were subsequently developed by Albert Einstein providing a physical explanation for the observed motion and establishing its connection to the microscopic properties of fluids [1].

Einstein's theoretical description relates the mean squared displacement of suspended particles to the diffusion coefficient and elapsed time [2]. This relationship provides a way to investigate the properties of a fluid through the motion of particles suspended within it. In particular, the Stokes-Einstein relation relates the diffusion coefficient to the particle size, fluid viscosity and temperature [3]. Consequently, by tracking the Brownian motion of particles with known properties, the viscosity of the surrounding fluid can be determined.

In this experiment, Brownian motion is used to determine the viscosity of aqueous suspensions containing monodisperse polystyrene microspheres of different sizes. By tracking the displacement of individual particles over time, their diffusion coefficients can be determined and related to the viscosity of the surrounding fluid. The main objective is to investigate whether this method provides a reliable determination of viscosity and to examine how the measured viscosity depends on the size of the suspended microspheres. If successfull, this would allow an *in situ*[4] measurement of the viscosity of the liquid. This is especially advantageous since it allows for measurements without disturbing the microsystem [5].

II. THEORY

In Brownian motion we assume that the motion of each distinct particle is independent of the motion of the other suspended particles. If we then also impose the condition that the number of particles per unit volume ν is a function of only x and t, i.e. $\nu = f(x,t)$, we get that it sattisfies the equation

$$\frac{\partial f}{\partial t} = D\frac{\partial^2 f}{\partial x^2}, \tag{1}$$

where D is the diffusion coefficient [2].
We then define $f(x,t)dx$ to denote the number of particles that have moved by a displacement between x and $x+dx$ in the x direction in the time interval $[0,t]$. We can impose the initial conditions as

$$f(x,0) = 0,$$
$$\int_{-\infty}^{\infty} f(x,0)dx = n,$$

where n is the total number of particles suspended in the liquid [2]. The solution to $f(x,t)$ can in this case be found analytically and is given by

$$f(x,t) = \frac{n}{\sqrt{4\pi Dt}} e^{-\frac{x^2}{4Dt}}. \tag{2}$$

We can then determine the average displacement of a particle along the x direction as

$$\lambda_x = \sqrt{\langle x^2 \rangle} = \sqrt{2Dt}, \tag{3}$$

where $\bar{x^2}$ is the arithmetic mean of squares of displacements along the x direction [2].
In the 2D case, we take the effect to be independent in the x and y axes. This then gives us an expression for the mean squared displacement in the xy-plane as

$$\langle R^2 \rangle = \langle x^2 \rangle + \langle y^2 \rangle = \langle x^2 + y^2 \rangle = 2Dt + 2Dt = 4Dt. \tag{4}$$

We can then use this relation to find an equation for the diffusion coefficient as

$$4Dt = \langle R^2 \rangle, \tag{5}$$
$$\frac{d}{dt}4Dt = \frac{d}{dt}\langle R^2 \rangle, \tag{6}$$
$$4D = \frac{d}{dt}\langle R^2 \rangle, \tag{7}$$
$$D = \frac{d\langle R^2 \rangle}{4dt}. \tag{8}$$

To find the viscosity using the diffusion coefficient we use the Stokes-Einstein relation for spherical particles, given as

$$D = \frac{k_B T}{6\pi\mu a}, \tag{9}$$

where k_B is the Boltzmann constant, T is the thermodynamic temperature, μ is the dynamic viscosity and a

is the particle radius. We can then get an expression for the dynamic viscosity as

$$\mu = \frac{k_B T}{6\pi D a} = \frac{2k_B T}{3\pi \frac{d\langle R^2 \rangle}{dt} a}. \tag{10}$$

Calculating the uncertainty is done using the formula

$$\sigma_A = A \cdot \sqrt{\sum_n \sigma_n^2}, \tag{11}$$

where A represents the quantity we want to calculate the resulting ucnertainty for and σ_n are relative uncertainties of quantities that are in the formula for A.

III. METHODOLOGY

As samples to be analysed, we are using monodisperse polystyrene microspheres of the sizes of $0.5 - 2\mu$m who are suspensed in water [6]. The resulting liquid is a collodial suspension, which is the description for mixtures where the collodial particles are significantly larger then the solvent molecules but small enough to show brownian motion [7]. In our case, the particles are dissolved towards concentration significantly lower then the value in which these disolvants are being sold, namely of 2.5% solids (w/v) aqueous suspension [6]. We have used a concentration of 0.1% across all measurements to minimize effects of differing concentration.
To calculate the final viscosity from Eqn. 10 we need to determine the temperature and diffusion coefficient. For determining the temperature we used a room thermometer located withing the room we were doing the experiment in. This thermometer had a precision of $0.1°C$. We calculate the diffusion coefficient by measuring the mean squared displacement $\langle R^2 \rangle$ of particles over time. For this we take a video of our particles using the DinoCapture 2.0 software. We then use the python library trackpy.py to track the movements of the particles recorded in the video. For the tracking to work correctly, the values of several parameters needed to be tailered to each video. An estimate the size of the particles in pixels, called the "feature size", was needed. This was moslty done uniformly across videos of particles with the same diameter. In some cases, however, the focus of the microscope made it so the value had to be tweaked for trackpy to track the particles correctly. We identified this as an artifact of diffraction within our particles where a bright spot would appear in the midle of the particle due to interference. This meant a higher feature size was needed to track the whole particle and not create behaviour where the tracking focused on only a part of the particle. Another important parameter was the so called "minmass". This is a measure of how strong the particle visibility needs to be for it to be tracked. By setting a higher minamss, the particles have to be clearer and in the foreground to be tracked. This helped eliminate accidentally tracking shadows of particles out of focus as well as any dust present in the setup.
To transfer the tracked mean square displacement from pixels to micrometers we needed to measure a calibration factor of how many micrometers was one pixel. For this we used a software called FiJi and a micrometer ruler that we placed under the microcsope with the same magnification as our videos. We then took a picture and measured the distance between stubs on the ruler in pixels, while we knew the physical distance between them. This yielded a calibration factor of $0.130 \pm 0.003 \mu m/px$.
To get the diffusion coefficient from this mean squared dispalcement calculation we used the formula in Eqn. 8. We made a linear fit through the values of mean squared displacement over time to get a slope (derivative), which was $4D$. We then just divided by 4 to get the diffusion coefficient. To help the linear fit get better data, we incormporated the initial condition that the mean squared dispalcement at time 0 is 0.

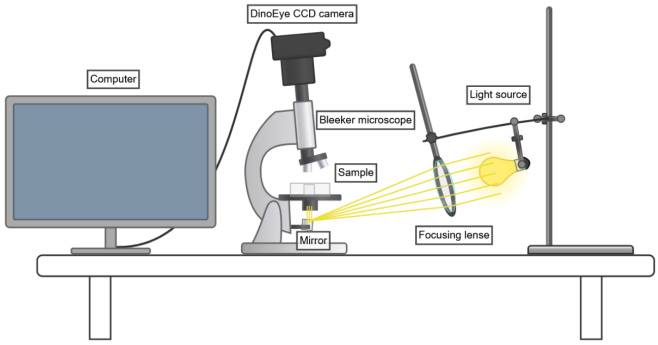


FIG. 1. Schematic showing our experimental setup.

IV. RESULTS

Our meausred value of temperature was of the order of $\sim 21°C$. The diffusion coefficient was of the order $\sim 10^{-12}$ and $\sim 10^{-13}$ depending on the particle radius. In the end we calcualted 25 viscosity values, 6 for $r = 0.375$, 6 for $r = 0.5$, 8 for $r = 0.75$

V. DISCUSSION

Discussion

VI. CONCLUSIONS

Conclusion

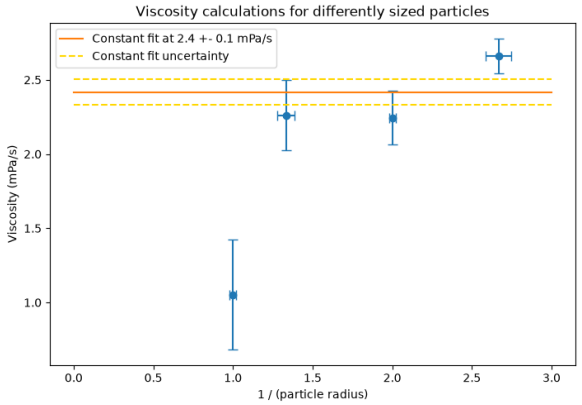


FIG. 2. The final result of average measurement of viscosity for particle diameters $d \in 2, 1.5, 1, 0.75\mu m$ and a fit of a cosntant function that yields resulting viscosity of $\nu = 2.4 \pm 0.1 mPa/s$.

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